

Macroeconomic Fundamentals and Cryptocurrency Prices: A Common Trend Approach*

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ABSTRACT

Based on asset pricing theory, we posit and find that equity markets and cryptocurrency markets share a common fundamental. Our cointegration tests show that the most important asset pricing primitive, consumption, can serve as the common fundamental. We further show that additional macroeconomic factors, as well as uncertainty and sentiment, all play a role in explaining the deviation from fundamentals. To understand the linkage between equity markets, cryptocurrency markets, and the macroeconomy, we suggest the following three channels: (i) portfolio allocation decisions, (ii) intermarket order flows, and (iii) technological adaption expectations.

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ABSTRACT

Based on asset pricing theory, we posit and find that equity markets and cryptocurrency markets share a common fundamental. Our cointegration tests show that the most important asset pricing primitive, consumption, can serve as the common fundamental. We further show that additional macroeconomic factors, as well as uncertainty and sentiment, all play a role in explaining the deviation from fundamentals. To understand the linkage between equity markets, cryptocurrency markets, and the macroeconomy, we suggest the following three channels: (i) portfolio allocation decisions, (ii) intermarket order flows, and (iii) technological adaption expectations.

1. Introduction

The novelty and volatility of cryptocurrency prices have led policymakers and investors to adopt the belief that its price dynamics are determined purely by speculative behavior from irrational agents.¹ This has spurred the fintech literature's efforts in trying to document stylized facts and theoretically understand the mechanisms that drive these empirical regularities. One of the key facts that has seemingly been uncovered is the lack of connection between equity markets and cryptocurrency markets. However, this is puzzling from an asset pricing perspective that predicts commonality between different markets.

We resolve this puzzle by using cointegration tests to show that cryptocurrency prices are indeed determined by economic fundamentals and investigate the role that the macroeconomy may have in driving the price process of cryptocurrencies. In contrast to conventional wisdom, our results show that cryptocurrency prices and equity prices are cointegrated and determined by common fundamentals. We find that cryptocurrency price contains a more transitory component reflecting the high volatility of cryptocurrencies compared to equity prices. The deviation from these fundamentals is not only due to speculative sentiment but also macroeconomic factors and time-varying uncertainty. These results rule out models where cryptocurrency prices are solely driven by "irrational exuberance".

Our study is the first, to our knowledge, in taking this approach to establish the long-run equilibrium relation between prices of equities, cryptocurrencies, and macroeconomic fundamentals, and examine whether macroeconomic fundamentals explain the fluctuations of the cryptocurrency market. Our methods and results complement Bhambhwani, Delikouras, and Korniotis (2019), who examine cointegration from a micro-perspective. We empirically investigate whether a long-run equilibrium exists between equity and cryptocurrency prices and if macroeconomic fundamentals drive the long-run equilibrium by taking a theoretically grounded empirical approach.

The theoretical motivation for a common long-run fundamental is relatively straightforward. The present value model implies that price is cointegrated with its fundamentals if the fundamental growth rate is stationary (see, e.g., Campbell & Shiller, 1987; Engel & West, 2005). However, the lack of consensus on cryptocurrency fundamentals makes it difficult to empirically identify. We circumvent this impediment by exploiting a nascent development in the asset pricing literature showing that return variation across asset classes can be explained by a common risk factor (e.g., Cochrane, 2011; Kojien, Moskowitz, Pedersen, & Vrugt, 2018;

¹For example, Jamie Dimon (CEO of JP Morgan) describes Bitcoin as a "fraud" and the traders investing in Bitcoin as "stupid" ("Jamie Dimon Slams Bitcoin as a 'Fraud'," 2017) while former Federal Reserve Chairman Alan Greenspan states "You have to really stretch your imagination to infer what the intrinsic value of Bitcoin is. I haven't been able to do it. Maybe somebody else can." ("Greenspan Says Bitcoin a Bubble Without Intrinsic Currency Value," 2013). The IMF notes that cryptocurrencies, unlike other financial assets "[have] their value depends entirely on self-fulfilling expectations" (IMF, 2019, p. 25).

Lettau, Maggiori, & Weber, 2014). Our study is also motivated by a key insight from Bossaerts (1988). In economies where two-fund separation holds in an approximate sense, Bossaerts (1988) argues that asset prices are cointegrated due to nonstationary common state variables.²

To address whether macroeconomic factors play a role in equilibrium, we take a two-fold approach. In our first approach, from the perspective of consumption-based asset pricing models, we consider an economy in which aggregate wealth consists of two asset classes: equities and cryptocurrencies. Motivated by Campbell and Mankiw (1989) and Lettau and Ludvigson (2001a), we focus on the representative agent's budget constraint and derive a cointegration relation between aggregate consumption, equities, and cryptocurrencies. This allows us to directly test for a common fundamental among these three variables. Our results strongly support the existence of a long-run equilibrium relation and indicate that consumption is able to serve as a common fundamental between equity and cryptocurrency prices. Our results are also consistent with the international finance literature that links currency returns to consumption growth (e.g., Burnside, 2012; Lustig, Roussanov, & Verdelhan, 2011).

In our second approach we investigate the determinants of deviations from the long-run equilibrium since the deviation predicts expected future cryptocurrency returns. Explaining the source of equilibrium deviations will help us understand the economic contents of cryptocurrency price fluctuations. We use asset pricing theory to help guide the set of candidate state variables: the consumption-wealth ratio, GDP growth, unemployment rate, uncertainty, and investor sentiment. Our empirical results show that macroeconomic fundamentals, uncertainty, and investor sentiment are important factors that influence cryptocurrency prices.

To further understand the long-run equilibrium relationship, we explore three plausible mechanisms linking the equity market, cryptocurrency market, and the macroeconomy. Firstly, the state of the economy jointly affects wealth, consumption, and investment decisions. Therefore, if investors hold cryptocurrency in their portfolios, their allocation decisions will jointly affect cryptocurrency and equity prices through demand changes driven by economic fluctuations. Secondly, there could be inter-market order flow in and out of the stock market and the cryptocurrency market related to the business cycle, analogizing from Hau and Rey (2006)'s exchange rate model. Finally, economic growth and technological innovations positively affect expectations of future technological adaptations and induce commonality (and therefore cointegration) between cryptocurrency and equity prices.

Our paper is most closely related to a recent study in Bhambhwani et al. (2019). Bhambhwani et al. (2019) is the first study to examine cointegration from a micro-perspective, i.e., between cryptocurrency price and a candidate list of cryptocurrency-related technological fundamentals. We both study the cointegration between cryptocurrency price and fundamentals and provide empirical evidence that there exists a

²In economies where two-fund separation holds strictly (e.g., CAPM) asset prices will be collinear with the state variables.

long-run equilibrium relation between cryptocurrency price and fundamentals. The two papers, however, differ in several dimensions.

First, Bhambhwani et al. (2019) study the cointegration between cryptocurrency price and plausible computational fundamentals (e.g., computing power and network size) from a micro perspective while we study the cointegration between cryptocurrency and macroeconomic state variables (aggregate consumption) from a macro perspective. Second, Bhambhwani et al. (2019) examine whether cryptocurrency returns are exposed to fundamental-based risk factors related to aggregate computing power and network size while we explore whether the deviation of the long-run relation is associated with macroeconomic fundamentals, aggregate uncertainty, and market sentiment. Third, we use the cointegration between cryptocurrency price and equity price as further evidence in support for commonality across asset classes.

We view our paper as a complementary to Bhambhwani et al. (2019) in that we are studying a similar question but with differing perspectives (micro vs macro). In short, our theoretically grounded results show that cryptocurrency prices and equity prices share macroeconomic fundamentals, consistent with macroeconomic-based asset pricing models and rule out models where cryptocurrency prices are driven by purely speculative sentiment.

Our results also contribute to various streams of the literature. In finance, an emerging literature is arising looking at cryptocurrencies from the trading and asset pricing perspective (e.g., Jain, McInish, & Miller, 2019; Pagano & Sedunov, 2020).³ In particular, Liu and Tsyvinski (2021) find that cryptocurrency returns are exposed to cryptocurrency network factors but not cryptocurrency production factors, and both momentum and public attention, help to explain the time series of cryptocurrency returns. However, there has been scant attention paid to whether more primitive asset pricing fundamentals impact cryptocurrency prices and link it to other asset markets.

Studies have also looked at the correlation between cryptocurrency prices and traditional asset classes such as stock, bonds, and commodities, failing to find any evidence of commonality (e.g., Baur, Hong, & Lee, 2018; Estrada, 2017). However, they neglect to take cointegration into account. Our paper aims to fill these gaps by applying a more appropriate framework (cointegration) and relating the macroeconomy to cryptocurrency price fluctuations. In addition to the main contributions we make to the fintech literature, our results shed light on the on-going debate in the exchange rate literature about whether exchange rates are related to macroeconomic fundamentals and equity markets (e.g., Engel & West, 2005; Hau & Rey, 2006; Lettau et al., 2014) and the literature on joint cross section returns of multiple asset classes (e.g., Cochrane, 2011; Koijen et al., 2018; Lettau et al., 2014).

The paper is organized as follows. Section 2 presents an illustrative model that derives the long-run equilibrium relations between the price of equity markets, cryptocurrency markets, and macroeconomic fundamentals. Section 3 outlines

³See Halaburda, Haeringer, Gans, and Gandal (2020) for a comprehensive survey of the extant literature.

and discusses our empirical methodology. Section 4 presents the empirical results. Section 5 examines the robustness of the results. Section 6 explores the potential mechanisms that give rise to cointegration. Section 7 concludes.

2. The Long-run Relation Between Equity Price, Cryptocurrency Price, and Fundamentals

Asset pricing theory illustrates that the key macroeconomic variables affecting the investment opportunity set should be priced factors in equilibrium (e.g., Breeden, 1979; Merton, 1973). Lettau et al., 2014 show that the cross-sectional variation in returns across asset classes can be captured by a common risk factor. The empirical evidence consistently suggests that the prices of equity, bond, and foreign currency appear to vary with macroeconomic variables (e.g., Engel & West, 2005; Fama & French, 1989; J. H. Stock & Watson, 1999). However, the prior literature has not found any evidence of correlation between cryptocurrency prices, equity prices, and macroeconomic variables.

To analyze the joint behavior of equity prices, cryptocurrency prices, and macroeconomic variables, we present a framework based on the budget constraint of a representative economy in which aggregate wealth is assumed to contain two types of assets: equity and cryptocurrency. Our framework is motivated by Campbell and Mankiw (1989) and Lettau and Ludvigson (2001a).

Let's consider an economy where W_t denotes the aggregate wealth of a representative agent in period t , C_t denotes his consumption in period t , and $R_{w,t+1}$ denotes the return on his aggregate wealth between periods t and $t + 1$. By studying the intertemporal budget constraint of the agent,

$$W_{t+1} = R_{w,t+1}(W_t - C_t). \quad (1)$$

Campbell and Mankiw (1989) derive the log consumption-wealth ratio and show that log consumption-wealth ratio is associated with expected future returns and consumption growth.

To mitigate the empirical obstacles that the unobservable component in aggregate wealth presents, Lettau and Ludvigson (2001a) extend the bivariate cointegration relation between consumption and wealth to a trivariate cointegration relation in consumption, assets, and labor income, and derive the log consumption, asset, and labor income ratio as follows,

$$c_t - \omega a_t - (1 - \omega)y_t = \mathbb{E}_t \sum_{j=1}^{\infty} \rho^j [\omega r_{w,t+j} + (1 - \omega)r_{h,t+j} - \Delta c_{t+j}] + (1 - \omega)z_t, \quad (2)$$

where c_t denotes log consumption, a_t denotes log asset wealth, y_t denotes labor income, $r_{w,t+1}$ denotes the log return on wealth, $r_{h,t+1}$ denotes the log return to human capital, Δc_{t+j} denotes log consumption growth, ρ is the discount factor, ω equals

the average share of asset holdings in total wealth, and z_t is a mean zero stationary random variable that allows the empirical linkage between human capital (which is unobservable) and labor income.

The left-hand side of equation (2) is denominated cay . Lettau and Ludvigson (2001a) demonstrate that c_t , a_t , and y_t share a common stochastic trend and cay is the cointegration residual. cay shows striking forecasting power for excess returns on aggregate stock market indexes. In addition, cay , as a superior summary measure of conditional expectation of latent state variable, helps explain the cross section of average stock returns (see, e.g., Lettau & Ludvigson, 2001b).

Our interest in this paper is to examine whether equity prices and cryptocurrency prices are associated with macroeconomic fundamentals. Following the spirit of Lettau and Ludvigson (2001a), we assume that aggregate wealth includes two classes of assets, equity and cryptocurrency while ignoring human capital. This assumption intends to incorporate cryptocurrency and equity holdings in the budget constraint to investigate whether equity price and cryptocurrency price share a long-run equilibrium relation with aggregate consumption rather than providing a flawless description of reality. In this case, the log aggregate wealth w_t can be approximated as linear combination of log equity price a_t and log cryptocurrency price b_t ⁴

$$w_t \approx \omega_a a_t + \omega_b b_t, \quad (3)$$

where ω_a is the equity weight and $\omega_b = (1 - \omega_a)$ is the cryptocurrency weight. With some simple algebra, we can derive the log consumption wealth ratio as follows,⁵

$$c_t - \omega_a a_t - \omega_b b_t = \mathbb{E}_t \sum_{j=1}^{\infty} \rho^j [\omega_a r_{a,t+j} + \omega_b r_{b,t+j} - \Delta c_{t+j}], \quad (4)$$

Equation (4) shows that the consumption, equity price and cryptocurrency price are cointegrated since the returns of equity, cryptocurrency, and consumption growth are presumed stationary. This cointegration indicates that equity price, cryptocurrency price, and consumption have a long-run equilibrium relation and that the prices of equities and cryptocurrencies share a common stochastic trend with consumption. The deviation from the equilibrium, which we denote as cab , predicts expected future equity returns and cryptocurrency returns if expected consumption growth is relatively stable.

3. Empirical Methodology

Equation (4) implies that the cointegration error can predict either future equity returns, cryptocurrency returns, and/or future consumption growth. If consumption,

⁴Our setup assumes that the shares of both assets are constant. This rather strong assumption is relatively innocuous because our interest is in understanding the long-run relationship between equity and cryptocurrency prices with consumption in a stylized setting.

⁵As is standard, the linearization constant is ignored throughout the paper without loss.

equity prices, and cryptocurrency prices are cointegrated and the three variables can be represented as a non-stationary p -order vector autoregression (VAR), then the Granger Representation Theorem (GRT) states that the vector error correction model (VECM) representation can be used to empirically estimate the predictive relationship.

Cointegration is based on the idea that a linear combination of the nonstationary variables will be covariance stationary if there exists a long-run equilibrium relationship that links the variables together. To present our cointegration framework formally, we consider a p -order VAR process,

$$\mathbf{x}_t = \boldsymbol{\nu} + \mathbf{A}_1 \mathbf{x}_{t-1} + \cdots + \mathbf{A}_p \mathbf{x}_{t-p} + \mathbf{e}_t. \quad (5)$$

where $\mathbf{x}_t$ denotes a vector of the variables of interest in period t , $\mathbf{A}_p$ is a matrix of the autoregressive coefficients for period $t - p$, $\boldsymbol{\nu}$ is a constant, and $\mathbf{e}_t$ is the residual vector. For example, $\mathbf{x}_t = [c_t, a_t, b_t]$ in our trivariate VAR (TVAR) model. One may consider setting up a TVAR using the first difference of $\mathbf{x}_t$ since the levels are nonstationary. However, this formulation is misspecified if $\mathbf{x}_t$ is cointegrated meaning that the moving average representation (MAR) of the first difference of $\mathbf{x}_t$ is noninvertible and no finite TVAR exists (see, e.g., Engle & Granger, 1987).

Using the GRT, we can show that the VAR(p) can be inverted to recover the VECM representation of the price process where a finite TVAR model exists,

$$\Delta \mathbf{x}_t = \boldsymbol{\alpha} \boldsymbol{\beta}' \mathbf{x}_{t-1} + \sum_{i=1}^{p-1} \boldsymbol{\Theta}_i \Delta \mathbf{x}_{t-i} + \mathbf{e}_t, \quad (6)$$

where $\boldsymbol{\alpha}$ is a vector containing the speed of adjustment coefficients that inform us which variable subsequently adjusts back to the common trend when deviation occurs; $\boldsymbol{\beta}$ is a vector containing the cointegration parameters; $\boldsymbol{\Theta}_i$ is a matrix of autoregressive coefficients; and $\boldsymbol{\beta}' \mathbf{x}_{t-1}$ represents last period's equilibrium error, or cointegrating residual, which measures the deviation from the long-run equilibrium relation and is stationary. Equation (6) is the model we use to empirically estimate the theoretical relationship derived in equation (4).

We first test for cointegration using the procedures in Johansen (1988, 1991, 1995). Johansen's procedures provide two test statistics, λ_{trace} and λ_{max} , for testing cointegration. The λ_{trace} statistic tests the null hypothesis that the number of distinct cointegrating vectors is equal to or less than r against a general alternative, while the λ_{max} statistic tests the null hypothesis that the number of cointegrating vectors is r against the alternative of $r + 1$ cointegrating vectors. The test procedures depend on the number of lags assumed in the VAR representation. We choose the optimal lags in the VAR based on Akaike's information criterion (AIC). We then empirically specify and estimate the VECM model using the maximum likelihood methods

derived by Johansen (1988, 1991, 1995).

4. Results

4.1. Features of the Data

We want to test the relationship between equity prices, cryptocurrency prices, and the macroeconomy. Therefore, we extract daily data for the S&P 500 price series from CRSP and the Bitcoin price series from CoinMarketCap as representative proxies of their respective markets and aggregate to a monthly series by using the end-of-month prices. We use data from Coinmetrics.io to calculate the market-cap weighted average of the ten largest cryptocurrencies as a proxy for the overall cryptocurrency market in our robustness tests. The sample consists of monthly data with a sample period of July 2010 until December 2021. We log transform and standardize the price series in order to ease interpretation and minimize matrix invertibility issues given the large magnitudes in differences between their prices (Kaufmann & Juselius, 2013).

Our macroeconomic variables come from a variety of sources. We use Markit's monthly GDP index. The data on *cay* are available at a quarterly frequency from Lettau's website and is available until 2019Q3. Following Guo and Whitelaw (2006), we interpolate the data into a monthly frequency. We use monthly personal consumption expenditure data from the US Bureau of Economic Analysis downloaded from FRED. Monthly unemployment rates and CPI-U from the US Bureau of Labor statistics are also downloaded from FRED. Our data on uncertainty come from Jurado, Ludvigson, and Ng (2015), Ludvigson, Ma, and Ng (2019), and S. R. Baker, Bloom, and Davis (2016). We also use the change in the CBOE Volatility Index to proxy for expected time-varying risk. The data on investor sentiment comes from two sources. The first source is a sentiment measure that has been orthogonalized with respect to macroeconomic measures (M. Baker & Wurgler, 2006) and runs until December 2018. The second source of sentiment data is the University of Michigan Survey of Consumer Sentiment accessed from FRED. All variables are standardized.

Figure 1 presents the times series of equity and Bitcoin log prices. We can see the expected nonstationarity of the price series and visually inspect the correlation in their fluctuations. From these results, there is suggestive evidence of a cointegrating relationship between nonstationary prices.

Table 1 presents summary statistics of log equity returns, log Bitcoin returns, and log consumption growth. The high volatility of Bitcoin relative to the equity market is obvious. We can also see large fluctuations across the board during 2020 due to COVID-19, which created economic turbulence reflected in extreme realizations for a number of macroeconomic and financial series for the US. In our study, this can be seen readily in the standard deviation of consumption. In 2020, there was a 56 b.p. standard deviation in monthly consumption growth, which is extremely high. Looking at annualized consumption growth, we find that consumption growth

experienced a 17% and 9% decline in April and May, respectively. However, the annualized consumption growth for 2020 excluding these two months ranged from about -4% to 4.5%. These extreme realizations will have strong effects on parameter estimates and forecasts generated by VECMs. Therefore, following the recommendation of Schorfheide and Song (2021), we exclude these outliers from the coming analysis.

Cointegration requires that the linear combination of nonstationary time series variables be stationary. As a preliminary step, we formally investigate the time-series properties of equity price, cryptocurrency price, and consumption. Table 2 presents the results from unit root tests on the level and difference of all the variables using both the modified Dickey-Fuller test (J. Stock, Elliott, & Rothenberg, 1996) and the Phillips and Perron (1988) test. The results fail to reject the null that there is a unit root for the equity price, cryptocurrency price, and consumption; and rejects the null for the changes of these variables. These results are indicative that equity price, Bitcoin price and consumption are possibly integrated of order 1.⁶ Therefore, we conduct cointegration tests and estimate a vector error correction model (VECM).

4.2. *Cointegration between Consumption, Equity Prices, and Bitcoin Prices*

We use Johansen's procedure to estimate the number of cointegrating relationships, which provides two tests for cointegration. We report both the "Trace" statistic and "L-max" statistic in Panel A of Table 3. Both "Trace" statistic and "L-max" statistic reject the null hypothesis of no cointegration at a significance level of at least 5% while failing to reject the null hypothesis that the cointegration rank is equal to or less than 1 and equal to 2 or less than 2. These results provide strong evidence of a single cointegrating relation among aggregate consumption, equity price, and cryptocurrency price. The existence of cointegration suggests that consumption and the price levels of both equity market and Bitcoin market have a long-run equilibrium relation and share a common stochastic trend, which establish direct evidence that the prices of equity and cryptocurrency are determined by aggregate consumption.

We argue that the deviations from the shared trend among aggregate consumption, cryptocurrency price, and equity price summarize the expectations of future equity returns and cryptocurrency returns if expected consumption growth is relatively stable. This suggests that the deviations predict either cryptocurrency returns and/or equity returns. To examine this argument, we run a VECM with changes in consumption, equity price, and Bitcoin price and report the results in Panel B of Table 3. We focus on the relation between the estimated trend deviation and future differences of consumption, equity price, and Bitcoin price, i.e., α_c , α_a and α_b .

The estimate of α_c is -0.008 with standard error of 0.006 and the estimate of α_a is 0.001 with a standard error of 0.006. In contrast, the estimate of α_b is relatively larger

⁶We additionally run an augmented Dickey-Fuller test and the results are qualitatively the same. Results available upon request.

in magnitude and statistically significant, 0.019 with a standard error of 0.005. The small and insignificant estimates of α_c and α_a indicate that the cointegrating residual does not predict aggregate consumption growth and equity returns. This result shows that consumption and equity contain mainly a permanent component and play little role to restoring the equilibrium subsequently. By contrast, the large and precise estimates of α_b indicates that the cointegrating error predicts Bitcoin returns. These results suggest that deviations from the shared trend among consumption, equity price, and Bitcoin price are better described as transitory movements in Bitcoin price than as transitory movements in consumption or equity price.

In sum, we show that equity markets, cryptocurrency markets, and consumption share a common fundamental. Consumption serves as a fundamental for cryptocurrency price and when the system experiences a shock, cryptocurrencies adjust towards the fundamental.

4.3. *Understanding Deviations from the Equilibrium*

Having established evidence that there exists a cointegration between consumption, equity price, and Bitcoin price and that the cointegration error predicts cryptocurrency returns, in this section we turn to explaining the cointegration error, i.e., the deviation from the long-run equilibrium. The deviation from equilibrium, termed *cab*, may be due to agents in these markets being driven by speculative sentiment or due to a rational time-varying risk factors explanation. In order to disentangle the importance of non-rational and rational explanations, we consider three potential sources associated with the deviation: investor sentiment, uncertainty, and macroeconomic fundamentals.

The literature shows that investor sentiment may contain a market-wide component that is likely to influence prices on many securities in the same direction at the same time (e.g., Barberis, Shleifer, & Vishny, 1998; Shiller, 2000). Therefore, we use the investor sentiment index from M. Baker and Wurgler (2006) and the University of Michigan Consumer sentiment survey as proxies for sentiment.

It is well known that uncertainty is associated with business cycle and rises sharply in recessions (e.g., Bansal & Yaron, 2004; Bloom, 2009; Jurado et al., 2015; Ludvigson et al., 2019). We use monthly data on time-varying macroeconomic uncertainty from Jurado et al. (2015) and Ludvigson et al. (2019). Macroeconomic uncertainty is estimated at three forecast horizons: 1 month, 3 months, and 12 months. We run a principal components analysis on these different forecast horizons to reduce the dimensionality of the data while capturing the time-varying uncertainty. We also consider the economic policy uncertainty measure of S. R. Baker et al. (2016).⁷

⁷S. R. Baker et al. (2016) construct this index from three types of underlying components. The first and most flexible component quantifies newspaper coverage of policy-related economic uncertainty. The second component draws on reports by the Congressional Budget Office (CBO) that compile lists of temporary federal tax code provisions. The third component of our policy-related uncertainty index draws on the Federal Reserve Bank of Philadelphia's Survey of Professional Forecasters.

Additionally, we use the CBOE's Volatility Index to proxy for expected time-varying risk.

Finally, we use macroeconomic variables that reflect the state of the real economy. We choose variables related to employment, GDP growth, and inflation. We also use the log consumption-wealth ratio, *cay*, from Lettau and Ludvigson (2001a) as a proxy for macro fundamentals since *cay* has been widely shown as a proxy for price of risk or an important state variable in asset pricing models.

To explore the potential sources of deviations from the equilibrium, we first run univariate regressions of the cointegrating residual on each of the competing variables which have potential to explain the deviation from the long-run equilibrium. Then we estimate multivariate regressions which combine the variables together.⁸

In Table 4, we explore whether sentiment helps explain *cab* from our VECM. We report estimated coefficient, Newey-West standard error robust to heteroskedasticity and autocorrelation in parentheses (with an optimal bandwidth equal to 4 months), the number of observations, and the adjusted- R^2 . We can see that the estimated coefficient of the UM measure of sentiment is statistically significant in column (2). A one standard deviation increase of consumer sentiment is negatively associated with a 0.404 standard deviation change of the cointegrating residual. Based on the adjusted- R^2 , the UM measure of sentiment explains more of the variation in the cointegration error relative to the Baker-Wurgler sentiment measure (16% vs 11%). In column (3), the estimated coefficients from the multivariate model are both significant with an adjusted- R^2 of 38%. This result suggests that speculative sentiment helps explain the fluctuation of Bitcoin price.

We explore whether uncertainty is associated with *cab* and report the result in Table 5. In column (2), the coefficient of Economic Policy Uncertainty ($\mathcal{U}_{BBD}$) is positive and significant at the 10% level with an adjusted- R^2 of about 7%. In column (3), the coefficient of changes in VIX is negative and significant at the 1% level. In column (4), we present the estimated coefficients from the multivariate model. In this specification, both VIX and $\mathcal{U}_{BBD}$ measures are statistically significant at the 1% level and the adjusted- R^2 is about 15%. A one-standard deviation change in macroeconomic uncertainty is associated with a 0.147 standard deviation increase in the cointegration residual, a one-standard deviation change in economic policy uncertainty is associated with a 0.510 standard deviation increase in the cointegration residual, and a one-standard deviation change in the VIX shock is associated with a 0.200 decrease in *cab*.

Table 6 reports the regression results for the macroeconomic measures. Column (1) shows that there is a negative relationship between *cay* and *cab* which is significant at the 5% level, with a one-standard deviation increase in *cay* associated with a 0.362 standard deviation decrease in the cointegrating residual. In column (2), we can see that a one-standard deviation increase in the unemployment rate is associated

⁸It is worth to note that our empirical design explores the conditional distribution of deviations from the equilibrium given a state variable. The estimated coefficients are not meant to imply economic causality but rather to document statistical associations.

with significant and positive 0.579 standard deviation change in *cab*. This variable explains the most variation in the cointegrating residual, with an adjusted- R^2 of 27%. In column (5), we can see that *cay* and unemployment retain their statistically significance.

In sum, results from the prior section document that *cab* contains information about future cryptocurrency returns. Our results in this section show that macroeconomic variables, uncertainty, and sentiment all play important roles in explaining these deviations from the equilibrium. This evidence suggests that Bitcoin price fluctuations are not determined solely by sentiment but also associated with rational time-varying risk factors.

5. Robustness

Over the last few years, new cryptocurrencies have begun to gain popularity and started to become bigger in terms of market share relative to Bitcoin. In this section, we repeat the same cointegration and determinant tests from the prior section but use a market-cap weighted average of the ten largest cryptocurrencies as a proxy for the overall cryptocurrency market.⁹

We conduct the trivariate cointegration tests using Johansen procedure and report both "Trace" statistic and "L-max" statistic in Panel A of Table 7. Both the "Trace" statistic and "L-max" statistic reject the null hypothesis of no cointegration at a 5% significance level while they fail to reject null hypothesis that the cointegration rank is equal to or less than 1 and equal to 2 or less than 2.

Next, we run the VECM estimation with changes in consumption, equity price, and the cryptocurrency basket price and report the results in Panel B of Table 7. We again focus on the relation between the estimated trend deviation and future differences of consumption, equity price, and cryptocurrency price, i.e., α_c , α_a and α_b . The estimate of α_c is -0.004 with standard error of 0.003 and the estimate of α_a is 0.001 with a standard error of 0.003, both estimates are small and insignificant. In contrast, the estimate of α_b is relatively larger in magnitude and statistically significant, 0.011 with a standard error of 0.003. The small and insignificant estimates of α_c and α_a indicate that the cointegrating residual does not predict aggregate consumption growth and equity returns. It is the price of cryptocurrency that takes action to restore the deviation back to the equilibrium. These results are in-line with the evidence provided in the previous section.

To explore the potential sources of deviations from the equilibrium, we conduct the same tests and estimations as in Tables 4, 5, and 6 but using a market-cap weighted average of the ten largest cryptocurrencies as a proxy for the overall cryptocurrency market. Table 8 reports the regression results of *cab* on the sentiment measures. We report estimated coefficient, standard error robust to heteroskedasticity and autocorrelation in parentheses (with an optimal bandwidth equal to 4 months), the

⁹We'd like to thank an anonymous referee for this suggestion. We additionally include summary statistics for bond and forex markets in the appendix.

number of observations, and the adjusted- R^2 . In column (2), the coefficient of the UM measure of sentiment (S_{UM}) is negative and significant at the 1% level with an adjusted- R^2 of about 21%. In column (3), we present the estimated coefficients from the multivariate model. In this specification, both sentiment measures are statistically significant at the 1% level and the adjusted- R^2 is 43%. These results are consistent with the results in Table 4.

Table 9 reports the regression results of cab on the uncertainty measures. In column (2), the coefficient of Economic Policy Uncertainty ($\mathcal{U}_{NB}$) is positive and significant at the 10% level with an adjusted- R^2 of about 6%. In column (3), the coefficient of changes in VIX is negative and significant at the 1% level with an adjusted- R^2 of about 2%. In column (4), we present the estimated coefficients from the multivariate model. In this specification, both uncertainty measures are statistically significant at the 10% level or better and the adjusted- R^2 is 14%. These results are consistent with the results in Table 5.

Table 10 reports the regression results of cab on the macroeconomic measures. Column (1) shows that there is a negative relationship between cay and the cointegration error which is significant at the 5% level, with a one-standard deviation increase in cay associated with a 0.323 standard deviation decrease in the cointegrating residual. Column (2) shows that a one-standard deviation increase in the unemployment rate is associated with a 0.517 standard deviation increase in the cointegrating residual. In Column (5), we can see that cay and unemployment retain their statistical significance. These results are consistent with those in Table 6.

All of the results in this section provide robust evidence that cryptocurrency markets are cointegrated with equity markets and the macroeconomy. The cointegrating error is not solely driven by speculative sentiment.

6. Discussion

The results from the previous section show that (i) there exists a common fundamental between equity and cryptocurrency price and (ii) the deviations from the equilibrium are related to macroeconomic fundamentals, uncertainty, and investor sentiment.

These two results that have been established have important implications. Firstly, these findings imply that the price of cryptocurrencies itself depends on macroeconomic fundamentals and uncertainty. This key finding overturns the conventional wisdom that views cryptocurrency prices as being unmoored from the economy or depending solely on sentiment. Secondly, we provide evidence that cryptocurrency price is related to fundamental macroeconomic variables – cay and unemployment – as well as VIX, economic policy uncertainty, and sentiment. That is, these macroeconomic variables are all plausible candidates for the cryptocurrency fundamental in addition to the monetary fundamentals (store of value, fungibility, scarcity, etc.) and technological fundamentals (decentralized network, blockchain, etc.) that the

literature has focused on. Therefore, investors can use this additional information when making portfolio allocation decisions.

We propose the following mutually non-exclusive plausible channels to explain the linkages between the macroeconomy, Bitcoin prices, and equity prices. First, the framework we use to motivate our trivariate tests provides one possible explanation. The state of the economy jointly affects wealth, consumption, and investment decisions. Therefore, if investors hold cryptocurrency in their portfolios, their allocation decisions will jointly affect Bitcoin and equity prices through demand changes driven by the business cycle. This explanation is related to studies trying to understand the commonality across asset classes. (e.g., Cochrane, 2011; Koijen et al., 2018; Lettau et al., 2014). This joint allocation decision would induce comovement between equities, consumption, and cryptocurrencies.

The second mechanism is in a similar spirit of international portfolio choices. Hau and Rey (2006) show that exchange rates, equity prices, and equity portfolio flows are jointly determined under incomplete foreign exchange risk trading in an equilibrium model. This model in our setting would imply higher returns in equity market relative to the cryptocurrency market are associated with US dollar depreciation. Net equity flows into the cryptocurrency market are positively correlated with a cryptocurrency appreciation. These dynamics generate positive correlation between Bitcoin price and equity price.

For example, when an agent purchases a good denominated in BTC, he initiates the purchase of BTC for USD. This implies there is a straightforward correspondence between net capital inflows and outflows between the US economy and Bitcoin economy. Aggregate wealth covaries with the business cycle, causing the stock price to positively covary with business cycle. If the US economy is in a state of expansion, net inflows into the Bitcoin economy will increase causing appreciation in BTC resulting in a positive correlation between the equity markets and the crypto market. While somewhat controversial at present, this order flow channel will grow in importance as the Bitcoin economy (the buying and selling of goods using Bitcoin) develops and grows.

A further possible mechanism is the interaction between the expectation of technological adaption and economic growth (Beaudry & Portier, 2006; Galor & Tsiddon, 1997; Pástor & Veronesi, 2009). This channel shows that prices incorporate forward-looking behaviors (rational and irrational). We argue that economic growth is related to future technological adaptations and induces cointegration between Bitcoin and equity prices.

7. Conclusion

We empirically establish that an equilibrium relationship exists between the equity market and cryptocurrency market. A simple model is formulated based on prior asset pricing theory where both markets are linked through common state variables. We then show that cryptocurrency price fluctuations are not due solely to speculative

sentiment, but also due to uncertainty and the state of the macroeconomy. These results are not only interesting from an academic perspective, but also contributes to the current contentious debate among regulators about how similar cryptocurrencies are to other assets and securities. This model is empirically supported and vindicated by the recent turmoil in equity and crypto markets in April and May of 2022. For example, the IMF notes that cryptocurrencies, unlike other financial assets "[have] their value depends entirely on self-fulfilling expectations" (IMF, 2019, p. 25). Our results are evidence that these virtual assets, like other more traditional assets, obey asset pricing theory and behave in similar ways to financial assets.

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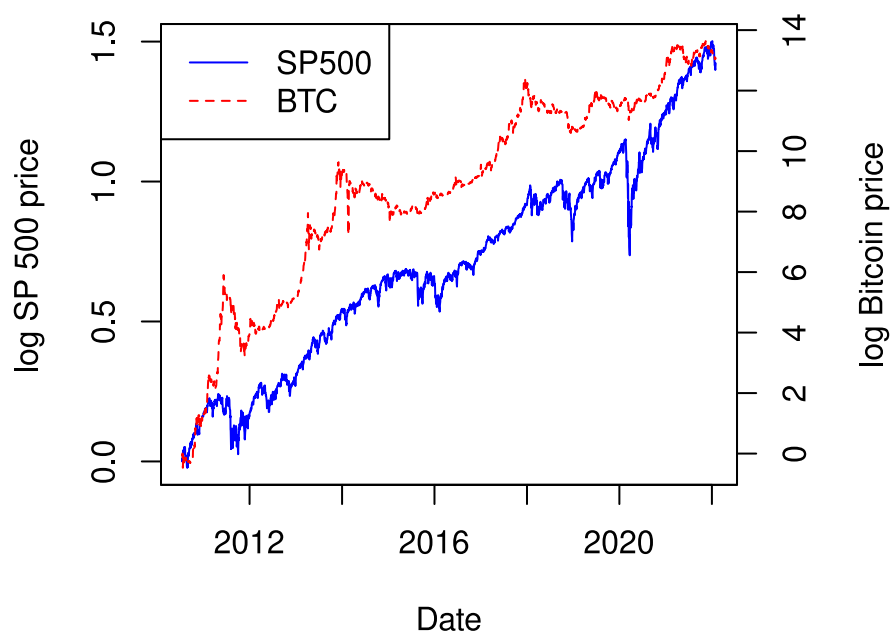


Figure 1. Price Series. This figure plots the daily log price level of the equity market (proxied by S&P500) and the cryptocurrency market (proxied by BTC). The sample period is from July 30, 2010 to December 31, 2021.

Table 1
Descriptive Statistics

The table shows summary statistics of the main variables: log real consumption growth (Δc), S&P 500 index log returns (Δa), and Bitcoin log returns (Δb). We present the mean (μ), the median (m), and the standard deviation (σ). The sample consists of monthly data with a sample period of July 2010 until December 2021.

	N	Consumption Δc_t (in bps)			S&P 500 Δa_t (in %)			Bitcoin Δb_t (in %)		
		μ	m	σ	μ	m	σ	μ	m	σ
2010	6	1.41	1.94	1.35	2.65	3.62	5.29	31.31	9.15	47.76
2011	12	0.44	0.72	1.37	-0.20	-0.51	4.72	19.09	-8.95	63.26
2012	12	0.96	0.76	2.84	0.76	1.25	3.04	8.20	7.21	14.02
2013	12	1.34	0.52	2.00	1.91	2.33	2.40	33.42	26.21	60.72
2014	12	3.96	3.99	1.77	1.31	1.89	1.93	-9.83	-13.88	21.60
2015	12	2.07	2.25	1.19	0.22	0.05	3.96	5.61	7.16	13.52
2016	12	2.04	1.83	2.82	1.30	0.27	2.35	8.41	9.04	11.84
2017	12	2.33	2.28	2.03	1.45	1.15	1.15	24.56	20.88	23.21
2018	12	1.34	2.28	3.38	-1.08	0.43	4.29	-9.13	-6.76	21.82
2019	12	1.54	1.88	1.95	1.62	2.02	3.46	6.72	7.26	22.40
2020	12	-3.43	3.23	56.35	1.38	3.64	7.85	10.29	8.63	21.67
2021	12	2.43	1.98	15.45	2.29	2.58	3.17	2.89	-1.86	24.30
Total	138	1.64	1.85	16.24	1.01	1.74	3.95	9.66	6.62	33.26

Table 2
Unit Root Tests

The table presents the results from a modified Dickey–Fuller t test for a unit root (Panel A) and the Phillips and Perron (1988) unit root test (Panel B). The modified Dickey–Fuller test is essentially an augmented Dickey–Fuller (ADF) test where the variable has been demeaned via a generalized least squares (GLS) regression before performing the test as proposed by Elliott, Rothenberg, and Stock (1996). The Phillips–Perron test corrects the ADF test for serial correlation. The null hypothesis for both of these tests is that there is a unit root. The alternative is that the variable was generated by a stationary process. The (1%, 5%, and 10%) critical values for the test in Panel A are (-2.58, -1.94, -1.62) and (-19.85, -13.73, -11.02) for the test in Panel B. Information criteria identify the optimal lag as 1-month while the Schwert criteria identifies 4-months as the optimal lag. Therefore, we provide results on the unit root tests up to 4 months. We use *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels. *Reads: The results show that all the levels are $I(1)$ and the differences are $I(0)$. This implies that that a linear combination of these variables might be cointegrated.*

Lags	Consumption		S&P 500		Bitcoin	
	c_t	Δc_t	a_t	Δa_t	b_t	Δb_t
Panel A: Modified Dickey-Fuller Unit Root Test						
1	-0.18	-10.89***	2.20	-4.09***	0.87	-6.11***
2	0.42	-8.69***	2.44	-2.92***	0.74	-5.68***
3	0.60	-7.70***	2.54	-2.20**	0.94	-4.94***
4	0.80	-7.05***	2.51	-1.93*	0.91	-4.97***
Panel B: Phillips-Perron Unit Root Test						
1	-3.52	-129.55***	-0.50	-0.50***	-155.51	-100.40***
2	-2.94	-116.08***	-0.42	-0.42***	-150.27	-102.52***
3	-2.46	-105.44***	-0.36	-0.36***	-145.37	-102.36***
4	-2.16	-98.73***	-0.32	-0.32***	-141.77	-101.88***

Table 3
Trivariate Estimations

This table presents estimates from the Johansen procedure in Panel A and the trivariate VECM in Panel B. The Johansen procedure provides two tests for cointegration. Under the null hypothesis, H_0 , that there are exactly r cointegrating relations, the “Trace” statistic supplies a likelihood ratio test of H_0 against the alternative, H_a , that there are k cointegrating relations, where k is the total number of variables in the model. A second approach uses the “L-max” statistic to test the null hypothesis of r cointegrating relations against the alternative of $r + 1$ cointegrating relations. We estimate a trivariate vector error correction model with one lag ($p = 2$ estimated via AIC),

$$\Delta \mathbf{x}_t = \alpha \boldsymbol{\beta}' \mathbf{x}_{t-1} + \sum_{i=1}^{p-1} \boldsymbol{\Theta}_i \Delta \mathbf{x}_{t-i} + \mathbf{e}_t$$

where α is a vector containing the speed of adjustment coefficients that inform us which variable subsequently adjusts back to the common trend when deviation occurs; $\boldsymbol{\beta}$ is a vector containing the cointegration parameters; $\boldsymbol{\Theta}_i$ is a matrix of autoregressive coefficients; and $\boldsymbol{\beta}' \mathbf{x}_{t-1}$ represents last period’s equilibrium error, or cointegrating residual, which measures the deviation from the long-run equilibrium relation and is stationary.

Panel A: Johansen Test									
H_0	H_a	L-max Eigenvalue Test				Trace Test			
		Critical-Values			Test Statistic	Critical-Values			Test Statistic
		10%	5%	1%		10%	5%	1%	
$r = 0$	$r > 0^*$	19.77	22.00	26.81	22.64	32.00	34.91	41.07	38.62
$r \leq 1^*$	$r > 1$	13.75	15.67	20.20	12.07	17.85	19.96	24.60	15.99
$r \leq 2$	$r > 2$	7.52	9.24	12.97	3.92	7.52	9.24	12.97	3.92
Panel B: VECM Estimation									
	Constant	Δc_{t-1}	Δa_{t-1}	Δb_{t-1}	$\boldsymbol{\beta}' \mathbf{x}_{t-1}$	N	$\bar{R}^2$		
Δc_t	0.021** (0.010)	-0.266*** (0.088)	0.219** (0.093)	0.062 (0.098)	-0.008 (0.006)	134	0.098		
Δa_t	0.035*** (0.010)	-0.193** (0.086)	-0.104 (0.091)	0.101 (0.096)	0.001 (0.006)	134	0.098		
Δb_t	0.023*** (0.008)	-0.122 (0.074)	0.110 (0.078)	0.221*** (0.082)	0.019*** (0.005)	134	0.223		

Table 4
The Determinants of the Trivariate Relationship: Sentiment Measures

This table presents the results from regressing cab on sentiment measures. The variable S_{BW} is the M. Baker and Wurgler (2006) sentiment index that has been orthogonalized with respect to macroeconomic measures. The variable S_{UM} is the sentiment index from the University of Michigan Survey of Consumer Sentiment. All variables are demeaned and adjusted by their standard deviation. Standard errors robust to heteroskedasticity and autocorrelation are shown in parenthesis. We use *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels.

	(1)	(2)	(3)
Dependent Variable: $c_t + 2.41a_t - 4.34b_t$			
Intercept	0.069 (0.197)	0.007 (0.147)	0.067 (0.157)
S_{BW}	-0.384 (0.245)		-0.495*** (0.162)
S_{UM}		-0.404** (0.158)	-0.590*** (0.153)
N	102	136	102
$\bar{R}^2$	0.110	0.160	0.380

Table 5
The Determinants of the Trivariate Relationship: Uncertainty Measures

This table presents the results from regressing cab on uncertainty measures. We use a monthly index of macroeconomic uncertainty and denote it by $\mathcal{U}_{LN}$. The index measures the macroeconomic uncertainty by taking the first principal component of macroeconomic uncertainty, financial uncertainty, and real uncertainty at the 1, 3, and 12 month forecast horizons (Jurado, Ludvigson, & Ng, 2015; Ludvigson, Ma, & Ng, 2019). We also use the economic policy uncertainty of Bloom (2009), $\mathcal{U}_{BB}$. This index measures policy-related economic uncertainty using three components: (i) quantifies newspaper coverage of policy-related economic uncertainty, (ii) the number of federal tax code provisions set to expire, and (iii) disagreement among economic forecasters. VIX denotes the change in the CBOE Volatility Index. All variables are demeaned and adjusted by their standard deviation. Standard errors robust to heteroskedasticity and autocorrelation are shown in parenthesis. We use *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels.

	(1)	(2)	(3)	(4)
	Dependent Variable: $c_t + 2.41a_t - 4.34b_t$			
Intercept	0.000 (0.160)	0.016 (0.155)	-0.001 (0.158)	0.025 (0.147)
$\mathcal{U}_{LN}$	0.012 (0.055)			0.147* (0.080)
$\mathcal{U}_{BB}$		0.311* (0.165)		0.510*** (0.176)
VIX			-0.160*** (0.052)	-0.200*** (0.062)
N	136	136	136	136
$\bar{R}^2$	-0.010	0.070	0.020	0.150

Table 6
The Determinants of the Trivariate Relationship:
Macroeconomic Measures

This table presents the results from regressing cab on macroeconomic measures. All variables are demeaned and adjusted by their standard deviation. Standard errors robust to heteroskedasticity and autocorrelation are shown in parenthesis. We use *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels.

	(1)	(2)	(3)	(4)	(5)
	Dependent Variable: $c_t + 2.41a_t - 4.34b_t$				
Intercept	-0.004 (0.151)	0.031 (0.141)	-0.026 (0.153)	-0.005 (0.157)	0.026 (0.139)
cay	-0.362** (0.169)				-0.196** (0.084)
Unemploy. Rate		0.579*** (0.195)			0.533*** (0.189)
dGDP			0.056 (0.042)		0.061 (0.065)
dCPI				0.145* (0.082)	0.013 (0.088)
N	136	136	136	136	136
$\bar{R}^2$	0.090	0.270	0.000	0.010	0.290

Table 7
Robustness: Trivariate Systems using a Basket of Cryptocurrencies

This table presents estimates from the Johansen procedure in Panel A and the trivariate VECM in Panel B. The Johansen procedure provides two tests for cointegration. Under the null hypothesis, H_0 , that there are exactly r cointegrating relations, the “Trace” statistic supplies a likelihood ratio test of H_0 against the alternative, H_a , that there are k cointegrating relations, where k is the total number of variables in the model. A second approach uses the “L-max” statistic to test the null hypothesis of r cointegrating relations against the alternative of $r + 1$ cointegrating relations. We estimate a trivariate vector error correction model with one lag ($p = 2$ estimated via AIC),

$$\Delta \mathbf{x}_t = \alpha \boldsymbol{\beta}' \mathbf{x}_{t-1} + \sum_{i=1}^{p-1} \boldsymbol{\Theta}_i \Delta \mathbf{x}_{t-i} + \mathbf{e}_t$$

where α is a vector containing the speed of adjustment coefficients that inform us which variable subsequently adjusts back to the common trend when deviation occurs; $\boldsymbol{\beta}$ is a vector containing the cointegration parameters; $\boldsymbol{\Theta}_i$ is a matrix of autoregressive coefficients; and $\boldsymbol{\beta}' \mathbf{x}_{t-1}$ represents last period’s equilibrium error, or cointegrating residual, which measures the deviation from the long-run equilibrium relation and is stationary.

Panel A: Johansen Test									
H_0	H_a	L-max Eigenvalue Test				Trace Test			
		Critical-Values			Test Statistic	Critical-Values			Test Statistic
		10%	5%	1%		10%	5%	1%	
$r = 0$	$r > 0^*$	19.77	22.00	26.81	22.89	32.00	34.91	41.07	39.04
$r \leq 1^*$	$r > 1$	13.75	15.67	20.20	12.15	17.85	19.96	24.60	16.15
$r \leq 2$	$r > 2$	7.52	9.24	12.97	4.00	7.52	9.24	12.97	4.00
Panel B: VECM Estimation									
	Constant	Δc_{t-1}	Δa_{t-1}	Δb_{t-1}	$\boldsymbol{\beta}' \mathbf{x}_{t-1}$	N	$\bar{R}^2$		
Δc_t	0.020** (0.010)	-0.267*** (0.088)	0.213** (0.093)	0.087 (0.089)	-0.004 (0.003)	134	0.010		
Δa_t	0.035*** (0.010)	-0.192** (0.086)	-0.105 (0.091)	0.097 (0.088)	0.001 (0.003)	134	0.099		
Δb_t	0.024*** (0.009)	-0.151* (0.080)	0.101 (0.084)	0.241*** (0.081)	0.011*** (0.003)	134	0.230		

Table 8
Robustness: Sentiment Determinants

This table presents the results from regressing cab estimated using a basket of the top-10 cryptocurrencies on sentiment variables. The variable S_{BW} is the M. Baker and Wurgler (2006) sentiment index that has been orthogonalized with respect to macroeconomic measures. The variable S_{UM} is the University of Michigan consumer sentiment index. All variables are demeaned and adjusted by their standard deviation. Standard errors robust to heteroskedasticity and autocorrelation are shown in parenthesis. We use *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels.

	(1)	(2)	(3)
	Dependent Variable: $c_t + 5.31a_t - 8.07b_t$		
(Intercept)	0.062 (0.201)	0.008 (0.142)	0.061 (0.151)
S_{BW}	-0.357 (0.249)		-0.481*** (0.152)
S_{UM}		-0.466*** (0.156)	-0.661*** (0.146)
N	102	136	102
$\bar{R}^2$	0.090	0.210	0.430

Table 9
Robustness: Uncertainty Determinants

This table presents the results from regressing cab estimated using a basket of the top-10 cryptocurrencies on uncertainty variables. We use a monthly index of macroeconomic uncertainty and denote it by $\mathcal{U}_{LN}$. The index measures the macroeconomic uncertainty by taking the first principal component of macroeconomic uncertainty, financial uncertainty, and real uncertainty at the 1, 3, and 12 month forecast horizons (Jurado, Ludvigson, & Ng, 2015; Ludvigson, Ma, & Ng, 2019). We also use the economic policy uncertainty of Bloom (2009), $\mathcal{U}_{NB}$. This index measures policy-related economic uncertainty using three components: (i) quantifies newspaper coverage of policy-related economic uncertainty, (ii) the number of federal tax code provisions set to expire, and (iii) disagreement among economic forecasters. VIX denotes the change in the CBOE Volatility Index. All variables are demeaned and adjusted by their standard deviation. Standard errors robust to heteroskedasticity and autocorrelation are shown in parenthesis. We use *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels.

	(1)	(2)	(3)	(4)
	Dependent Variable: $c_t + 2.41a_t - 4.34b_t$			
Intercept	0.000 (0.160)	0.015 (0.155)	-0.001 (0.158)	0.024 (0.148)
$\mathcal{U}_{LN}$	0.022 (0.057)			0.153* (0.080)
$\mathcal{U}_{BBD}$		0.290* (0.168)		0.493*** (0.179)
VIX			-0.152*** (0.050)	-0.188*** (0.063)
N	136	136	136	136
$\bar{R}^2$	-0.010	0.060	0.020	0.140

Table 10
Robustness: Macroeconomic Determinants

This table presents the results from regressing cab estimated using a basket of the top-10 cryptocurrencies on macroeconomic variables. All variables are demeaned and adjusted by their standard deviation. Standard errors robust to heteroskedasticity and autocorrelation are shown in parenthesis. We use *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels.

	(1)	(2)	(3)	(4)	(5)
	Dependent Variable: $c_t + 5.31a_t - 8.07b_t$				
(Intercept)	-0.003 (0.152)	0.028 (0.147)	0.000 (0.160)	-0.005 (0.157)	0.023 (0.146)
cay	-0.323** (0.154)				-0.165** (0.081)
Unemploy. Rate		0.517** (0.201)			0.478** (0.196)
dGDP			0.061 (0.043)		0.065 (0.064)
dCPI				0.141* (0.084)	0.025 (0.090)
N	136	136	136	136	136
$\bar{R}^2$	0.070	0.210	0.000	0.010	0.220

Table A1
Descriptive Statistics with Bond and Forex Markets

The table shows summary statistics of the main variables: log real consumption growth (Δc), S&P 500 index log returns (Δa), Bitcoin log returns (Δb), bond market returns (Δa^{bond}), and forex returns (Δa^{forex}). We present the mean (μ), the median (m), and the standard deviation (σ). The sample consists of monthly data with a sample period of July 2010 until December 2021.

	N	Consumption Δc_t (in bps)			S&P 500 Δa_t^{SP} (in %)			Bitcoin Δb_t (in %)			Bonds Δa_t^{bond} (in %)			Forex Δa_t^{forex} (in %)		
		μ	m	σ	μ	m	σ	μ	m	σ	μ	m	σ	μ	m	σ
2010	6	1.41	1.94	1.35	2.65	3.62	5.29	31.31	9.15	47.76	-0.38	-0.52	2.65	-0.78	-1.06	1.89
2011	12	0.44	0.72	1.37	-0.20	-0.51	4.72	19.09	-8.95	63.26	1.40	1.44	1.79	0.06	0.02	1.46
2012	12	0.96	0.76	2.84	0.76	1.25	3.04	8.20	7.21	14.02	0.24	-0.29	1.64	-0.05	0.18	1.29
2013	12	1.34	0.52	2.00	1.91	2.33	2.40	33.42	26.21	60.72	-0.51	-0.45	1.81	0.30	0.20	1.16
2014	12	3.96	3.99	1.77	1.31	1.89	1.93	-9.83	-13.88	21.60	0.56	0.44	0.98	0.77	0.24	1.14
2015	12	2.07	2.25	1.19	0.22	0.05	3.96	5.61	7.16	13.52	-0.28	-0.38	1.29	0.65	0.26	1.59
2016	12	2.04	1.83	2.82	1.30	0.27	2.35	8.41	9.04	11.84	-0.20	-0.14	1.85	0.06	0.27	1.65
2017	12	2.33	2.28	2.03	1.45	1.15	1.15	24.56	20.88	23.21	0.23	0.22	0.89	-0.58	-0.67	1.19
2018	12	1.34	2.28	3.38	-1.08	0.43	4.29	-9.13	-6.76	21.82	0.22	0.23	1.41	0.60	0.48	0.91
2019	12	1.54	1.88	1.95	1.62	2.02	3.46	6.72	7.26	22.40	0.66	-0.04	1.87	0.07	0.22	0.52
2020	12	-4.19	3.23	24.51	1.69	3.64	9.86	12.58	21.44	24.23	0.85	1.04	3.05	-0.67	-0.85	1.49
2021	12	2.43	1.98	15.45	2.29	2.58	3.17	2.89	-1.86	24.30	0.01	0.18	1.14	0.47	0.37	0.94
Total	138	1.67	1.85	8.17	1.03	1.61	4.12	9.80	6.62	33.52	0.30	0.08	1.75	0.12	0.20	1.34